

(T5)

$$\xi \sim R[\theta; 2\theta] \quad \Rightarrow \bar{x}_n$$

$$\Rightarrow \xi \sim S(x|\theta) = \frac{1}{\theta} \cdot \mathbb{I}_{(\theta, 2\theta)}$$

а) В)

• ОММ

Возьмем  $d_1$ : 
$$d_1 = \int_{-\infty}^{+\infty} x g dx = \int_{\theta}^{2\theta} \frac{x}{\theta} dx = \frac{3}{2} \theta$$

$$d_1 = \bar{X}_1 = \bar{x} \Rightarrow \frac{3}{2} \theta = \bar{x} \Rightarrow \underline{\tilde{\theta}} = \frac{2}{3} \bar{x} \quad - \text{ОММ}$$

• Несмещенность

$$M[\tilde{\theta}] = \frac{2}{3} M\bar{x} = \frac{2}{3} M\xi = \theta \Rightarrow \tilde{\theta} - \text{несмещенная}$$

• Состоятельность

$$D[\tilde{\theta}] = \frac{4}{9} D\bar{x} = \frac{4}{9n} D\xi = \frac{\theta^2}{27n} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \tilde{\theta} - \text{состоит}$$

$$* d_2 = \int_{\theta}^{2\theta} \frac{x^2}{\theta} dx = \frac{8-1}{3} \theta^2 = \frac{7}{3} \theta^2 \Rightarrow D\xi = \frac{7}{3} \theta^2 - \frac{9}{4} \theta^2 = \frac{\theta^2}{12} \neq 0$$

$\Rightarrow \tilde{\theta} - \text{состоит.}$

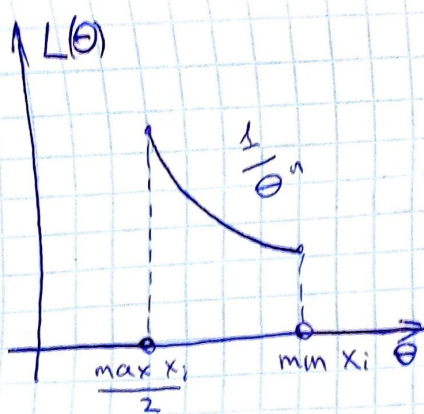
• ОМП

р.з. правдоподобия

$$L(\theta) = \prod_{i=1}^n S(x_i, \theta) = \frac{1}{\theta^n} \left\{ \forall i \rightarrow \begin{matrix} \min_{1 \leq i \leq n} x_i \geq \theta \\ \max_{1 \leq i \leq n} x_i \leq 2\theta \end{matrix} \right\} =$$

$$= \frac{1}{\theta^n} \left\{ \frac{\max x_i}{2} \leq \theta \leq \min x_i \right\}$$





$$\Rightarrow \sup_{\theta} L(\theta) \Rightarrow \underline{\hat{\theta} = \frac{\max x_i}{2}}$$

ОМП

• Несмещ.

$$M[\hat{\theta}] = \frac{1}{2} M[\max x_i] = \frac{1}{2} \int_0^{2\theta} t \frac{n}{\theta} \left(\frac{t}{\theta} - 1\right)^{n-1} dt \quad \textcircled{2}$$

$$/* (f(t))^n = \left(\frac{t-\theta}{\theta}\right)^n */ \quad /* u = \frac{t}{\theta} */$$

$$\textcircled{2} \frac{\theta}{2} \int_1^2 n u (u-1)^{n-1} du = \theta - \frac{\theta}{2n+2} \Rightarrow \frac{\theta(2n+1)}{2n+2}$$

- смещен  $\Rightarrow \hat{\theta}' = \frac{2n+2}{2n+1} \hat{\theta} = \frac{n+1}{2n+1} \max x_i$

- несмещ.

• Состоятельность

$$D[\hat{\theta}'] = M[\hat{\theta}'^2] - M^2[\hat{\theta}']$$

$$M[\hat{\theta}'^2] = \frac{1}{4} \int_0^{2\theta} t^2 \frac{n}{\theta} \left(\frac{t}{\theta} - 1\right)^{n-1} dt = \frac{\theta^2}{4} \frac{2(2n^2+4n+1)}{(n+2)(n+1)}$$

$$D[\hat{\theta}'] = \theta^2 \left( \frac{2n^2+4n+1}{2(n+2)(n+1)} - \left( \frac{2n+1}{2n+2} \right)^2 \right) =$$

$$= \theta^2 \frac{n}{4(n+1)^2(n+2)}$$

$$D[\hat{\theta}'] = \frac{4(n+1)^2}{(2n+1)^2} \frac{n}{4(n+1)^2(n+2)^2} \theta^2 \xrightarrow{n \rightarrow \infty} 0 \Rightarrow$$

$\Rightarrow \hat{\theta}'$  - состоятельна



$$c) D_1[\tilde{\theta}] = \frac{\theta^2}{27n}$$

$$D_2[\tilde{\theta}'] = \frac{n\theta^2}{(n+2)^2(2n+1)^2}$$

IV

$D_2[\tilde{\theta}']$  - более точная

$$d) ] f(\bar{x}_n, \theta) = \frac{\max_{i=1-n} x_i}{\theta} - 1 \Rightarrow$$

$$\Rightarrow P(f < t) = P(\max x_i < \theta t + \theta) = (F(\theta t + \theta))^n$$

$$\text{где } \begin{cases} 0 & , x < \theta \\ \frac{x}{\theta} - 1 & , \theta \leq x \leq 2\theta \\ 1 & , x > 2\theta \end{cases} \Rightarrow$$

$$\Rightarrow P(f < t) = \begin{cases} 0 & , t \leq 0 \\ t^n & , \theta \leq t \leq 1 \\ 1 & , t > 1 \end{cases}$$

$$t_1 : F(t_1) = \frac{1-\beta}{2}$$

$$t_2 : F(t_2) = \frac{1+\beta}{2}$$

$$\Rightarrow t_{1,2} = \sqrt[n]{\frac{1 \pm \beta}{2}}$$

$$\text{Тогда } P\left(t_1 < \frac{\max x_i}{\theta} - 1 < t_2\right) = \beta$$

$$\text{т.е. } t_1 + 1 < \frac{\max x_i}{\theta} < t_2 + 1$$

$$\Rightarrow P\left(\frac{\max x_i}{1 + \sqrt[n]{\frac{1+\beta}{2}}} < \theta < \frac{\max x_i}{1 + \sqrt[n]{\frac{1-\beta}{2}}}\right) = \beta$$

- точный довер. интер.

e)

• ОММ

$$\tilde{\theta} = \frac{2}{3} \bar{x} = g(d_1)$$

$$g(d_1) = \theta ; \bar{\nabla} g = g' = \frac{2}{3} ; K = d_2 - d_1^2$$



$$\frac{\sqrt{n}(\tilde{\theta} - \theta)}{\sqrt{\left(\frac{2}{3}\right)^2(\tilde{\alpha}_2 - \tilde{\alpha}_1^2)}} = \frac{3n(\tilde{\theta} - \theta)}{2\sqrt{S^2(n-1)}} \rightsquigarrow N(0, 1), \text{ где } S^2 = \frac{n}{n-1} \cdot \tilde{\mu}_2$$

т.к. норм. распред  $t_{12} = \pm 1,96$

$$P\left(t_1 < \frac{3n(\tilde{\theta} - \theta)}{2\sqrt{S^2(n-1)}} < t_2\right) = \beta$$

$$\Rightarrow \frac{2t_1\sqrt{S^2(n-1)}}{3n} < \tilde{\theta} - \theta < \frac{2t_2\sqrt{S^2(n-1)}}{3n}$$

$$\Rightarrow P\left(\frac{2}{3}\bar{x} - \frac{3,92\sqrt{S^2(n-1)}}{3n} < \theta < \frac{2}{3}\bar{x} + \frac{3,92\sqrt{S^2(n-1)}}{3n}\right) = \beta$$

- асимпт. довер. интер. ОММ